PIC 10A 1A

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Today…

- HW5 Hints
- Arrays Examples
  - Sorting with Functions

HW5 Hints

HW5 Hints

- 1. Get inputs
- 2. Compare Arrays
  - 3. If sizes are different, they are not equal
  - 4. If sizes are the same,
    - 5. Compare elements: if any of them in the same position are different, they are not equal
    - 6. Otherwise, they are equal
- 7. Print output

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HW5 Hints

- 1. Get inputs

- `cin.clear();` and `cin.ignore();` are necessary here, to clear the invalid input ('Q') remaining in the buffer

e.g. Buffer:

`[7][ ][3][ ][9][ ][Q]`

`cin` failed here and the buffer needs to be cleared

Reading Input

If you DON'T know how many input values the user will supply, then you add values to an array until the end of input has been reached, or when current_size exceeds CAPACITY.

```
double values[CAPACITY];
int current_size = 0;
double input;
while (cin >> input) {
    if (current_size < CAPACITY) {
        values[current_size] = input;
        current_size++;
    }

}
```

HW5 Hints (optional)

- 1. Get inputs
- If you implement a function for reading an array, it will make your code cleaner and easier to read
- e.g.

```
cout << "Enter array 1 (Q to quit): ";
size1 = read_inputs(arr1, MAX_SZ);
cout << "Enter array 2 (Q to quit): ";
size2 = read_inputs(arr2, MAX_SZ);
```

Functions Should Return or Update New Array Size

When a function changes the size of an array, it should indicate to the caller how many elements the array has after the call. One way to do this is to return the new size.

```
int read_inputs(double inputs[], int capacity) {
    int current_size = 0;
    double input;
    while (cin >> input) {
        if (current_size < capacity) {
            inputs[current_size] = input;
            current_size++;
        }
    }
    return current_size;
}
```

HW5 Hints

- 2. Compare Arrays: e.g. set a flag such as `bool isEqual`
  - 3. If *sizes are different*, the arrays are *not equal*
    - OK, this is easy!
  - 4. If *sizes are the same*,
    - 5. Compare elements: if any of them in the same position are different, they are not equal
      - Use a loop
      - As soon as different elements are found, _____ and break.
    - 6. *Otherwise*, they are equal
      - What is the good choice of the initial value of `isEqual`?
- 7. Print output (typical `cout` with an `if-else` clause)

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Array Examples

Sort and Search with Functions

Array Examples

- Sorting is covered in the lecture
- As an exercise for both functions and arrays, we will repeat it *using functions*

- Also, we will implement functions accepting array parameters which
  - 1. Reads input parameters
  - 2. Prints the subarray of index range [beg, end)
  - 3. Finds the index of the maximum element in the range [beg, end)
  - 4. Swaps two entries
  - 5. Sorts the array in *descending* order

- [Live Code]

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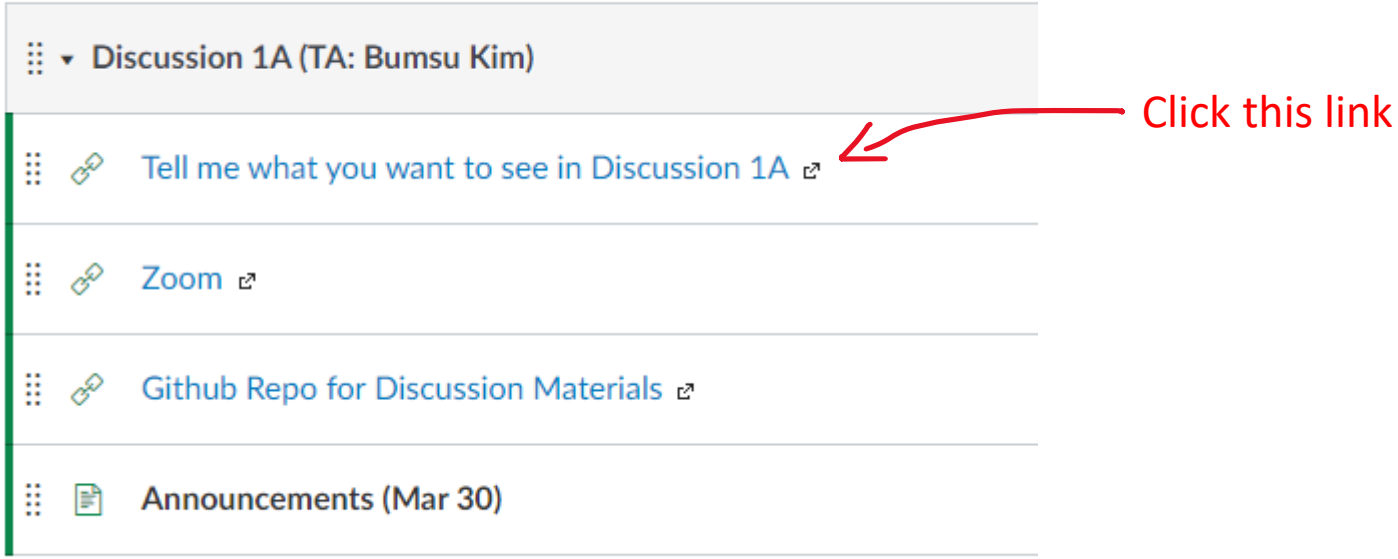
Your Feedback is welcome

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